



Tutorial 03 – Paraview scripting

We have been using Paraview through its Graphical User Interface (GUI). However, Paraview can also be acted upon programmatically (e.g., via Python scripts) and, in fact, it is possible to completely dispense with the GUI.

Tracing the GUI commands

Paraview facilitates the development of Python scripts by recording (tracing) all commands issued by the user. To start with do:

- Select **Tools → Start Trace**

Now, proceed with the sequence of commands presented in the next sub-section

Visualizing a static vector field

Our goal is to create a set of streamlines from a velocity vector field, and visualize the velocity as arrows (glyphs) at given points of the streamlines.

- Open Examples/disk_out_ref.ex2
- **Apply**



- Click the **-X** icon



- **Stream tracer** filter
- Verify that streamtracing is on V :



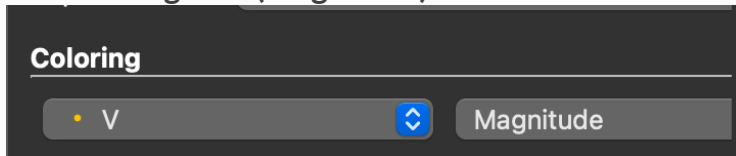
Vectors

• V

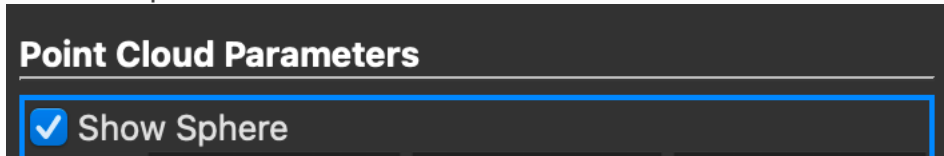


- Set **Seed Type** to **Point Cloud**.
- **Apply**

- Set **Coloring** to **V (Magnitude)**



- Change the Opacity to 0.6.
- Hide the sphere



- Zoom the Camera to be close to the data:



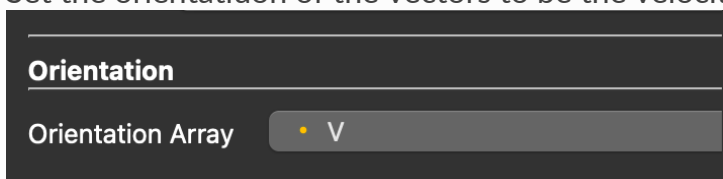
Let's filter the output such that the data refers to some value of the integration time as generated by the StreamFilter

- Select **Filters** → **Common** → **Contour** , or press
- Contour on **IntegrationTime**.
- **Apply**.

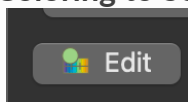


Let's associate an arrow (glyph) to each point output by the Countour Filter

- Select **Filters** → **Common** → **Glyph** , or press
- Set the orientatiuon of the vectors to be the velocity **V**



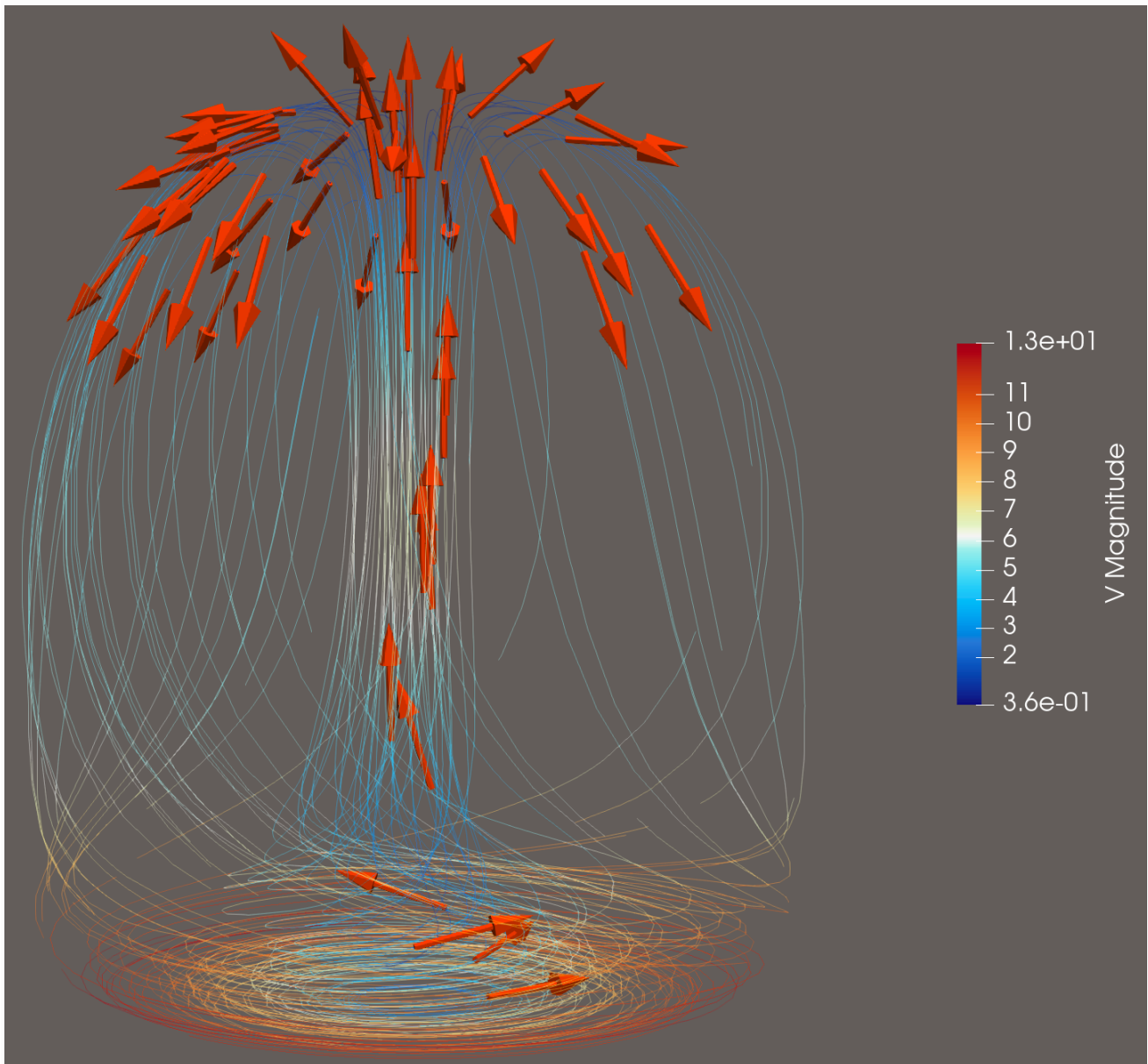
- No Scale Array.
- Scale 1.
- Set **Glyph Mode** to **All Points**.
- Set **Coloring** to **Solid Color** and choose a color that contrasts with the StreamLines



(use

- **Apply**.

Your view should be similar to the image below:



To terminate recording (tracing) user commands, do:

- Select **Tools → Stop Trace**

The Python Shell must have popped up. First step is to save this script

- Select **File → Save as Script...**

Navigate through the code, checking that each GUI command has a corresponding programmatic counterpart.

Scripts can be used both within and outside Paraview's GUI.

- Reset Paraview by pressing 
- Select **Tools → Python Script Editor**
- If the script is not loaded into the editor, load it using **File → Open...**
- Run the script by selecting **File → Run**

You should be able to see in Paraview's Render view the same visualization that has been previously generated interactively.

Running scripts

Scripts can be executed without using Paraview's GUI. `pvpython` is Paraview's Python command, allowing precisely this.

However, when using this execution mode, results have to be explicitly forwarded somewhere, such as a program for post-processing or a disk for storage.

- Edit the script you just saved and add the following line at the end:

```
myview = GetActiveView()
SaveScreenshot(<path>/<filename>'.png', myview)
```

Now you are ready to execute the script without resorting to the Paraview's GUI. From a command line, and making sure you are able to access `pvpython`, execute

- `pvpython <my-script>.py`

Verify the output image, which should be identical to the one previously created interactively.

Extra content

Generating an animation from the velocity static vector field

If you have a vector field in your data, you can animate a static dataset.

- Launch Paraview
- Run the script we have been working on (you can delete the screenshot saving lines if saving the image again is not desired)

This data set does not contain multiple time instants, but since V is a vectorial field the Stream Tracer filter, traces the velocity over a time interval. In fact, if you inspect the Contour Value Range you can see the time interval values:

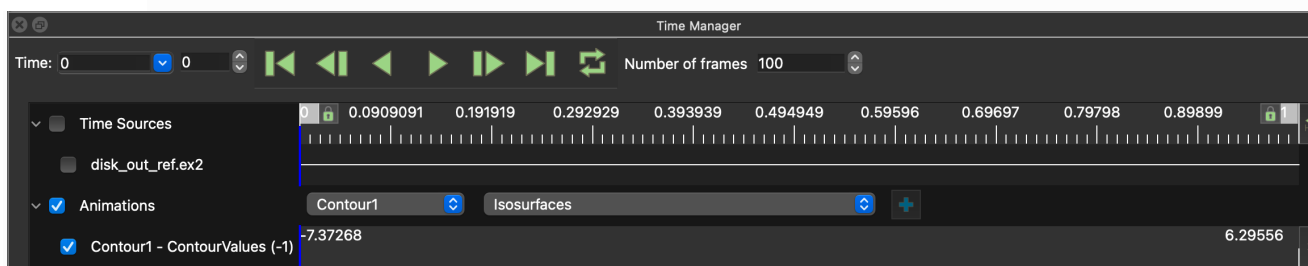
Isosurfaces

Value Range: [-7.37268, 6.29556]

So an animation of the glyphs (arrows) can be made by changing over time (animation time) the integration time the Contour Filter is passing onto the Glyph filter.

Let's open the Time Manager window:

- Select **View → Time Manager**.
- Since there are no time instants in the data uncheck **Time Sources**.
- Change the pulldown box next to **Animations** to be **Contour**.
- Click the blue **+**. Note that the time interval is set to be the same as the value range in the contour by time integration. Change the initial time to 0, just because the animation will look more interesting
- Set **No. Frames** to **150**.



- Now, click the play button.

Save this as a movie.

- **File → Save Animation.**
- Add a file name.
- Save as a **.avi**.
- **OK.**
- **OK.**